

Assignment -02 Programming Examples

Verilog Programming Exercises – Language Essentials

1. Largest of Three Numbers

Write a Verilog program to find the largest among three integer numbers.

2. Smallest of Five Numbers

Write a Verilog program to find the smallest among five integer numbers stored in an array.

3. Even or Odd

Write a Verilog program to determine whether a given integer is even or odd.

4. Positive, Negative, or Zero

Write a Verilog program to determine whether a given number is positive, negative, or zero.

5. Simple Calculator

Write a Verilog program to perform addition, subtraction, multiplication, division, and modulus operations using a case statement.

6. Multiplication Table

Write a Verilog program to print the multiplication table of a given number from 1 to 10.

7. Factorial of a Number

Write a Verilog program to calculate the factorial of a given integer using a loop.

8. Fibonacci Series

Write a Verilog program to generate the first N terms of the Fibonacci series.

9. Prime Number Test

Write a Verilog program to determine whether a given integer is a prime number.

10. Prime Numbers in a Range

Write a Verilog program to display all prime numbers between 1 and 100.

11. Reverse an Integer

Write a Verilog program to reverse the digits of a given integer.

12. Palindrome Number

Write a Verilog program to determine whether a given integer is a palindrome.

13. Sum of Digits

Write a Verilog program to calculate the sum of all digits of a given integer.

14. Armstrong Number

Write a Verilog program to determine whether a given three-digit number is an Armstrong number.

15. Array Statistics

Store 10 integers in an array and write a Verilog program to find the maximum, minimum, sum, and average.

16. Bubble Sort

Store 10 integers in an array and write a Verilog program to sort them in ascending order using the Bubble Sort algorithm.

17. Second Largest Element

Store 10 integers in an array and find the largest and second-largest elements without sorting the array.

18. Frequency of an Element

Store 10 integers in an array and write a program to count how many times a specified number occurs.

19. Binary Number Analysis

Given an 8-bit vector, write a Verilog program to count the number of 1s and 0s and determine whether the MSB and LSB are 1.

20. Student Marks Processing

Store the marks of 10 students in an integer array and calculate the total, average, highest mark, lowest mark, and number of students in different mark ranges.